

## Tutorial Quiz 4 Rubric

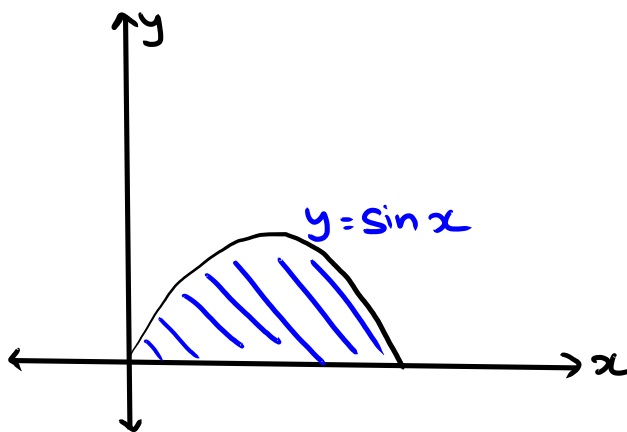
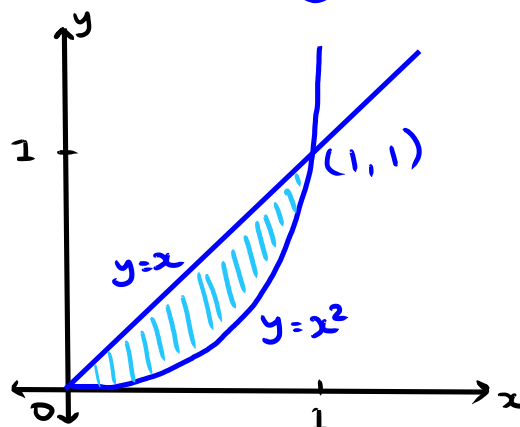
Q<sub>1</sub>]

$$\int_0^{\pi} \int_0^{\sin x} y \, dy \, dx$$

$$\Rightarrow \int_0^{\pi} \left[ \frac{y^2}{2} \right]_0^{\sin x} dx \Rightarrow \int_0^{\pi} \frac{1}{2} \sin^2 x \, dx$$

$$\Rightarrow \frac{1}{4} \int_0^{\pi} (1 - \cos 2x) \, dx$$

$$\Rightarrow \frac{1}{4} \left[ x - \frac{1}{2} \sin 2x \right]_0^{\pi/4} = \frac{\pi}{4} \quad - (1)$$

Q<sub>2</sub>] We begin by sketching the region:

Now using the first moment formula

$$M_x = \iint_R y \, dA$$

$$= \int_0^1 \int_{x^2}^x y \, dy \, dx$$

$$= \int_0^1 \left[ \frac{y^2}{2} \right]_{x^2}^x dx$$

$$= \int_0^1 \left( \frac{x^2}{2} - \frac{x^4}{2} \right) dx$$

$$= \left[ \frac{x^3}{5} - \frac{x^5}{10} \right]_0^1$$

$$= \frac{1}{15} \quad - (1)$$

$$M_y = \iint_R x \, dA$$

$$= \int_0^1 \int_{x^2}^x x \, dy \, dx$$

$$= \int_0^1 [xy]_{x^2}^x dx$$

$$= \int_0^1 (x^2 - x^3) dx$$

$$= \left[ \frac{x^3}{3} - \frac{x^4}{4} \right]_0^1$$

$$= \frac{1}{12} \quad - (1)$$

\*Note: it is entirely possible to have other ways to get to this answer one possibility is to merely choose a different order of integration

i.e.  $M_x = \int_0^1 \int_x^{x^2} y \, dx \, dy$